

Soal No.3 - k-NN (k = 3) - Laporan Lengkap

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Metode: k-NN (k=3)

Rumus yang digunakan (sesuai dosen):

- Normalisasi A1 (min-max): $A1' = (A1 - \min) / (\max - \min)$
- Encoding: yes = 1, no = 0 (A2-A6)
- Jarak Euclidean: $d = \sqrt{\sum (x_i - y_i)^2}$

1) Data Training (14 baris):

Idx | A1 | A2 A3 A4 A5 A6 | A7

1 | 35.9 | no yes no no no | no
2 | 36.0 | no no yes yes yes | yes
3 | 36.0 | no yes no no no | no
4 | 36.7 | no no yes yes yes | yes
5 | 36.7 | no yes no no no | no
6 | 36.7 | no yes no no no | no
7 | 37.0 | no no yes no no | yes
8 | 37.1 | no yes no no no | no
9 | 38.0 | no yes yes no yes | no
10 | 38.1 | no yes yes no yes | no
11 | 38.3 | no yes yes no yes | no
12 | 39.7 | no yes yes no yes | no
13 | 40.0 | yes yes yes yes yes | yes
14 | 40.2 | yes yes yes yes yes | yes

2) Normalisasi A1 (min=35.9, max=40.2):

35.9->0.000, 36.0->0.023, 36.7->0.186, 37.0->0.256, 37.1->0.279,
38.0->0.488, 38.1->0.512, 38.3->0.558, 39.7->0.884, 40.0->0.953, 40.2->1.000

3) Data Testing (5):

Test1: 40.0 yes yes no yes no

Test2: 37.9 no no yes no no

Test3: 38.0 no yes yes no yes

Test4: 37.3 no yes no no no

Test5: 37.3 no no yes yes yes

4) Vektor fitur (A1', A2..A6):

Test1: [0.953,1,1,0,1,0]

Test2: [0.465,0,0,1,0,0]

Test3: [0.488,0,1,1,0,1]

Test4: [0.372,0,1,0,0,0]

Test5: [0.372,0,0,1,1,1]

5) Tabel jarak lengkap (14 jarak) untuk setiap test (Idx, A1', A2..A6, Class, Distance):

--- Test1 ---

Idx	A1'	A2	A3	A4	A5	A6	Class	Distance
13	0.953488	1	1	1	1	1	Yes	1.414214

14	1.000000	1	1	1	1	1	Yes	1.414978
8	0.279070	0	1	0	0	0	No	1.566793
5	0.186047	0	1	0	0	0	No	1.609027
6	0.186047	0	1	0	0	0	No	1.609027
3	0.023256	0	1	0	0	0	No	1.692729
1	0.000000	0	1	0	0	0	No	1.705620
12	0.883721	0	1	1	0	1	No	2.001217
11	0.558140	0	1	1	0	1	No	2.038701
10	0.511628	0	1	1	0	1	No	2.048229
9	0.488372	0	1	1	0	1	No	2.053371
7	0.255814	0	0	1	0	0	Yes	2.118195
4	0.186047	0	0	1	1	1	Yes	2.142029
2	0.023256	0	0	1	1	1	Yes	2.205652

--- Test2 ---

Idx	A1'	A2	A3	A4	A5	A6	Class	Distance
7	0.255814	0	0	1	0	0	Yes	0.209280
2	0.023256	0	0	1	1	1	Yes	0.302274
4	0.186047	0	0	1	1	1	Yes	0.302274
9	0.488372	0	1	1	0	1	No	1.414433
10	0.511628	0	1	1	0	1	No	1.415021
11	0.558140	0	1	1	0	1	No	1.415984
1	0.000000	0	1	0	0	0	No	1.436392
3	0.023256	0	1	0	0	0	No	1.492526
5	0.186047	0	1	0	0	0	No	1.524662
6	0.186047	0	1	0	0	0	No	1.524662
8	0.279070	0	1	0	0	0	No	1.547984
12	0.883721	0	1	1	0	1	No	1.755605
13	0.953488	1	1	1	1	1	Yes	1.781901
14	1.000000	1	1	1	1	1	Yes	1.869123

--- Test3 ---

Idx	A1'	A2	A3	A4	A5	A6	Class	Distance
9	0.488372	0	1	1	0	1	No	0.000000
10	0.511628	0	1	1	0	1	No	0.023271
11	0.558140	0	1	1	0	1	No	0.069759
12	0.883721	0	1	1	0	1	No	0.397652
2	0.023256	0	0	1	1	1	Yes	1.243603
4	0.186047	0	0	1	1	1	Yes	1.264298
13	0.953488	1	1	1	1	1	Yes	1.414214
14	1.000000	1	1	1	1	1	Yes	1.414978
1	0.000000	0	1	0	0	0	No	1.443714
3	0.023256	0	1	0	0	0	No	1.499498
5	0.186047	0	1	0	0	0	No	1.531636
6	0.186047	0	1	0	0	0	No	1.531636

8 0.279070 0 1 0 0 0 No 1.553390
7 0.255814 0 0 1 0 0 Yes 1.566387

--- Test4 ---

Idx	A1'	A2	A3	A4	A5	A6	Class	Distance
8	0.279070	0	1	0	0	0	No	0.046488
5	0.186047	0	1	0	0	0	No	0.139523
6	0.186047	0	1	0	0	0	No	0.139523
3	0.023256	0	1	0	0	0	No	0.201305
1	0.000000	0	1	0	0	0	No	0.214243
7	0.255814	0	0	1	0	0	Yes	0.228023
4	0.186047	0	0	1	1	1	Yes	0.360602
2	0.023256	0	0	1	1	1	Yes	0.450249
9	0.488372	0	1	1	0	1	No	1.215455
10	0.511628	0	1	1	0	1	No	1.229807
11	0.558140	0	1	1	0	1	No	1.268708
12	0.883721	0	1	1	0	1	No	1.549391
13	0.953488	1	1	1	1	1	Yes	1.683522
14	1.000000	1	1	1	1	1	Yes	1.789047

--- Test5 ---

Idx	A1'	A2	A3	A4	A5	A6	Class	Distance
4	0.186047	0	0	1	1	1	Yes	0.139523
2	0.023256	0	0	1	1	1	Yes	0.302274
7	0.255814	0	0	1	0	0	Yes	1.415922
3	0.023256	0	1	0	0	0	No	1.422075
1	0.000000	0	1	0	0	0	No	1.438364
5	0.186047	0	1	0	0	0	No	1.697122
6	0.186047	0	1	0	0	0	No	1.697122
8	0.279070	0	1	0	0	0	No	1.709175
9	0.488372	0	1	1	0	1	No	2.001662
10	0.511628	0	1	1	0	1	No	2.012730
11	0.558140	0	1	1	0	1	No	2.047221
12	0.883721	0	1	1	0	1	No	2.332922
13	0.953488	1	1	1	1	1	Yes	2.434501
14	1.000000	1	1	1	1	1	Yes	2.524875

Summary Predictions (k=3):

Test1: Yes

Test2: Yes

Test3: No

Test4: No

Test5: Yes